# Tensile strength



**What is tension?**
Tension is a pulling force where forces are acting from opposite ends.



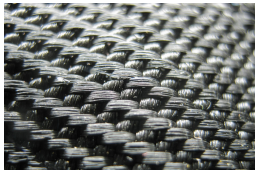
**Tensile strength can be summarised as** being the maximum pulling force a material can withstand before starting to deform or break.
(a material's ability to resist being pulled apart or stretched)

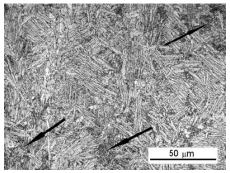
**When measuring a materials tensile strength theory look for**

- Ultimate tensile strength (UTS), represents the highest and max stress a material can withstand
- Yield strength, the stress when it starts to deform permanently
- Elongation (%), how much the material stretches before breaking also known as ductility

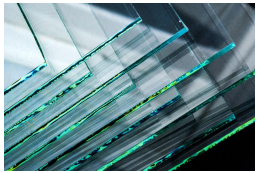
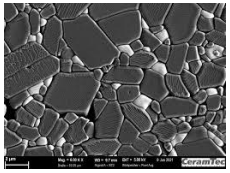
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Carbon based materials such as graphene and carbon fiber have great tensile strength due to its unique interlocking structure. This is the reason why its used in airplanes and f1 cars, withstanding huge pulling forces during liftoff as well as stopping parts from tearing at high speeds. Other materials with high tensile strength are tungsten and titanium alloys.






Materials with low tensile strength are usually brittle substances which fracture easily like concrete, glass, ceramics and some soft metals. Mainly due to a lack of internal bonds so can't resist stretching.

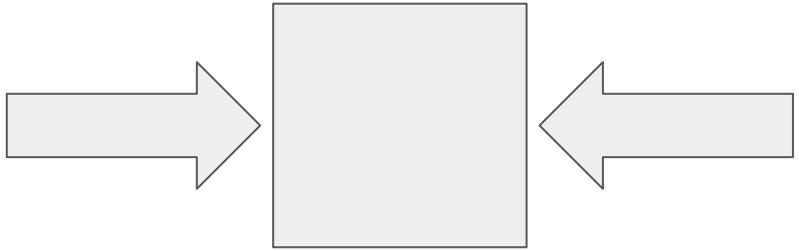




Tensile strength can be lowered ion high temperatures as it allows atoms to move more freely.

# Compressive strength



**Compressive strength** is a material's ability to resist being crushed or squeezed by pushing forces (compression) without deforming significantly or failing.



**Key factors which can affect compressive strength is the samples:**
**shape**- Different shapes distribute force differently. For example, a short, wide cylinder resists compression better than a tall, thin one, which may buckle or fail

https://youtu.be/p00nMx-ULIA?si=xuPJ9qG1-M6_4rpE

**Size**- Larger samples are more likely to contain small defects or inconsistencies, which can reduce measured compressive strength compared to smaller, more uniform samples

**Moisture content**- Higher moisture levels can weaken many materials (like concrete or wood) because water interferes with internal bonding

Materials which have low compressive strength can include plastics foam and rubber.
Rubber is a good example due to its clear compression when applied with force, being flexible and elastic (not rigid) and finally when under high pressure it simply squashes rather than resisting



A material that has very high compressive strength is **concrete**. This is because it can withstand very large loads without crushing, making it perfect for buildings and foundations. However it doesn't perform as well under tension.



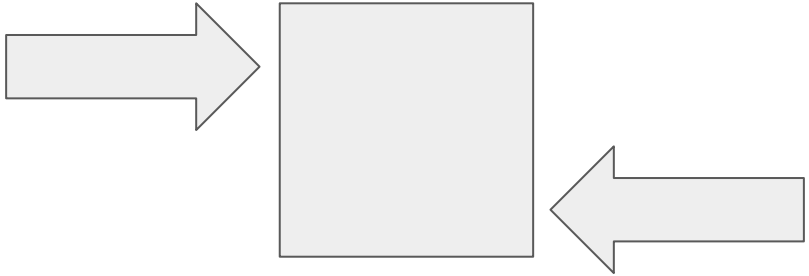




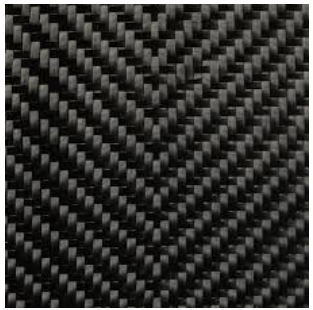
# Shear strength

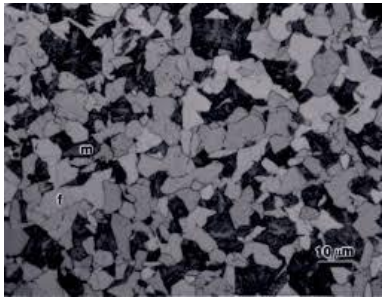


**Shear strength** is a material's ability to resist forces that cause layers within the material to slide past one another. Shear force is when a force pushes parts of a material in opposite directions, parallel to the surface)
Unlike tension or compression, shear causes the internal pieces to slip and move.



**Steel has a very high shear strength** due to its strong metallic bonds which help it to resist layers sliding past each other. This is why steel is used in bolts and rivets to secure materials together. It can withstand large sideways forces before failing.
more examples of materials that have similar properties are titanium and weaved carbon fibre.






**Wood** is a very good example of a material with low shear strength due to its uniform grain. Where the fibres are aligned they can easily slide apart and is much weaker (parallel) ,when across the grain it can withstand much more shear than before.



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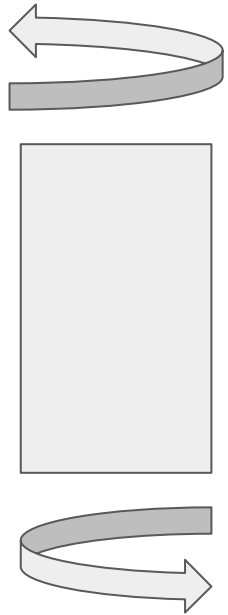
# Torsion strength



**Torsion strength** is a material's ability to resist twisting forces (torque) and it represents the max twisting stress it can handle before breaking.

There are many factors which can affect the materials torsion strength,
**The type of material** is important since products needing to withstand high levels of torque usually use steel due to its strong metallic bonds and structure. Whereas brittle materials like ceramics and cast iron can fail much quicker under torsion.

**Dimensions** such as length mean that longer components can twist more than shorter components when under the same level of torque, therefore having a higher risk of failing



**High temperatures** reduce torsional strength ( due to atoms moving more freely



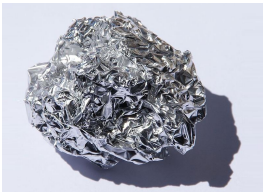
**Low temperatures** cause materials to become more brittle leading to sudden failures



Materials with **bad (low) torsional strength** are usually brittle or very flexible, meaning they crack or permanently deform when twisted

**Examples being:**
Soft metals like aluminum
Many plastics
Woods





But **high torsion strength materials** include:
Titanium alloys
Advanced composites (fiberglass)





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